

PRODUCT REQUIREMENTS DOCUMENT

NeuroSync AI

Behavioral Digital Twin for Predictive Health Monitoring

Version	1.0
Date	May 13, 2026
Owner	SEN SABU
Status	Active Development

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1. Product Overview

One-liner: NeuroSync AI is a cloud-native predictive healthcare platform that creates behavioral digital twins using wearable and lifestyle data to identify abnormal behavioral drift and generate explainable health-risk predictions.

Field	Detail
Product Name	NeuroSync AI
Owner	Sen Sabu
Institution	Lovely Professional University
Date	May 16, 2026
Version	1.0
Primary Language	Python, TypeScript, JavaScript
Primary Domain	AI-Powered Predictive Healthcare
Deployment Model	Cloud-Native
Architecture	Microservice-Based
Status	Academic Assessment Project

Description

NeuroSync AI is a full-stack AI healthcare analytics platform designed to monitor behavioral and physiological patterns using wearable-device data and lifestyle activity streams.

The platform continuously analyzes:

- sleep irregularities,
- sedentary duration,
- movement intensity,
- activity transitions,
- stress indicators,
- and behavioral drift patterns.

Machine learning models generate predictive health-risk insights while Explainable AI modules provide transparent explanations behind predictions.

The platform uses:

- Docker containerization,
- Kubernetes orchestration,
- Jenkins CI/CD pipelines,
- Terraform-based cloud provisioning,
- and scalable cloud infrastructure for deployment and monitoring.

2. Objectives and Goals

1. Develop a predictive healthcare analytics platform using AI/ML.
2. Create behavioral digital twins from wearable data streams.
3. Detect abnormal behavioral drift patterns.
4. Provide explainable health-risk predictions.
5. Demonstrate scalable cloud-native architecture.
6. Implement DevOps-based deployment workflows.
7. Support preventive healthcare monitoring through analytics dashboards.
8. Maintain modular and scalable system architecture.

3. Target Users

User Persona	Type	Description
Health-Conscious User	Primary	Tracks wellness and behavioral health
Research Student	Primary	Studies wearable healthcare analytics
Healthcare Analyst	Secondary	Reviews predictive behavioral insights
AI/ML Researcher	Secondary	Analyzes healthcare prediction models
System Administrator	Technical	Monitors infrastructure and deployment
DevOps Engineer	Technical	Manages CI/CD and cloud infrastructure

4. Actual Problem Statement

Core Problem: Most wearable healthcare applications focus only on basic activity tracking and fail to analyze long-term behavioral drift for proactive health-risk prediction.

Existing healthcare systems suffer from multiple limitations:

- Limited predictive analytics
- No behavioral digital twin modeling
- Lack of explainable AI insights

- Poor cloud scalability
- No integrated DevOps deployment workflow
- Minimal real-time behavioral monitoring

Current wearable applications often provide:

- isolated health metrics,
- simple dashboards,
- or fitness recommendations,

without intelligent behavioral analysis or scalable predictive infrastructure.

Specific Pain Points:

1. Lack of behavioral drift analysis
2. Poor explainability in healthcare AI systems
3. Limited integration of cloud-native analytics
4. No scalable deployment workflow for healthcare AI platforms
5. Fragmented wearable data processing systems
6. Difficulty monitoring long-term behavioral changes.

5. Proposed Solution

NeuroSync AI addresses these challenges through a multi-layer cloud-native architecture:

Layer 1 — Wearable & Data Collection Layer

- wearable device integration
- activity data collection
- sleep monitoring
- behavioral event logging

Layer 2 — Cloud Analytics Infrastructure

- cloud storage
- distributed data synchronization
- scalable API services
- behavioral data pipeline

Layer 3 — AI/ML Prediction Engine

- behavioral drift analysis
- anomaly detection
- predictive health scoring
- explainable AI visualization

Layer 4 — DevOps & Deployment Layer

- Docker containerization
- Jenkins CI/CD automation
- Kubernetes orchestration
- Terraform infrastructure provisioning

Why this works:

- scalable cloud-native architecture
- explainable predictive healthcare analytics
- modular AI pipeline
- automated deployment workflows
- production-style DevOps integration
- industry-oriented system design

6. Features

6.1 F1 — Behavioral Digital Twin Engine

- personalized behavioral profiles
- historical pattern analysis
- behavioral drift tracking
- dynamic health baselines

6.2 F2 — AI Risk Prediction System

- predictive wellness scoring
- anomaly detection
- fatigue prediction
- stress-pattern analysis

6.3 F3 — Explainable AI Dashboard

- prediction explanations
- feature importance visualization
- confidence scoring
- interpretable recommendations

6.4 F4 — Wearable Data Integration

- activity tracking
- sleep monitoring
- movement analysis
- behavioral-event collection

6.5 F5 — Cloud-Based Analytics Engine

- distributed cloud storage
- real-time synchronization
- scalable analytics processing
- API-driven architecture

6.6 F6 — DevOps Deployment

- automated CI/CD workflows
- Docker container builds
- Kubernetes deployments

- infrastructure automation using Terraform

6.7 F7 — Smart Health Dashboard

- wellness analytics
- trend visualization
- behavioral insights
- risk alerts

6.8 F8 — Authentication & Security

- JWT authentication
- encrypted records
- role-based access
- secure API communication

7. User Flow

7.1 Primary Flow — Predictive Health Monitoring

1. User registers and logs into NeuroSync AI
2. User connects wearable or uploads health data
3. Backend APIs process incoming behavioral data
4. Cloud storage synchronizes activity records
5. AI engine analyzes behavioral patterns
6. Behavioral digital twin updates continuously
7. Risk prediction engine identifies abnormalities
8. Explainable AI dashboard displays insights
9. Users receive wellness alerts and recommendations

7.2 Secondary Flow — DevOps Deployment Pipeline

1. Developer pushes code to GitHub
2. Jenkins pipeline triggers automatically
3. Docker images are built and tested
4. Kubernetes deploys updated services
5. Terraform provisions infrastructure resources
6. Monitoring system tracks deployment health

8. Functional Requirements

ID	Feature Requirement	Priority
FR-01	User registration and authentication	High
FR-02	Wearable data synchronization	High
FR-03	Behavioral drift analysis	High
FR-04	AI-based risk prediction	High
FR-05	Explainable AI visualization	High
FR-06	Dashboard analytics generation	Medium
FR-07	Cloud data storage	Medium
FR-08	CI/CD pipeline automation	Medium
FR-09	Kubernetes deployment support	Medium
FR-10	Secure session management	High
FR-11	Real-time wellness alerts	Medium
FR-12	Infrastructure provisioning using Terraform	Medium

9. Non-Functional Requirements

9.1 Performance

- Real-time behavioral analytics processing
- Dashboard load time below 2 seconds
- Scalable concurrent API handling

9.2 Security

- JWT-secured APIs
- Encrypted user health records
- Secure cloud communication
- Protected deployment secrets

9.3 Reliability

- Automated deployment rollback
- Fault-tolerant cloud architecture
- Continuous monitoring and logging

9.4 Usability

- Intuitive dashboard UI
- Simple wearable synchronization
- Explainable prediction visualization

9.5 Maintainability

- Modular microservice architecture
- Containerized services
- Independent deployment workflows

9.6 Portability

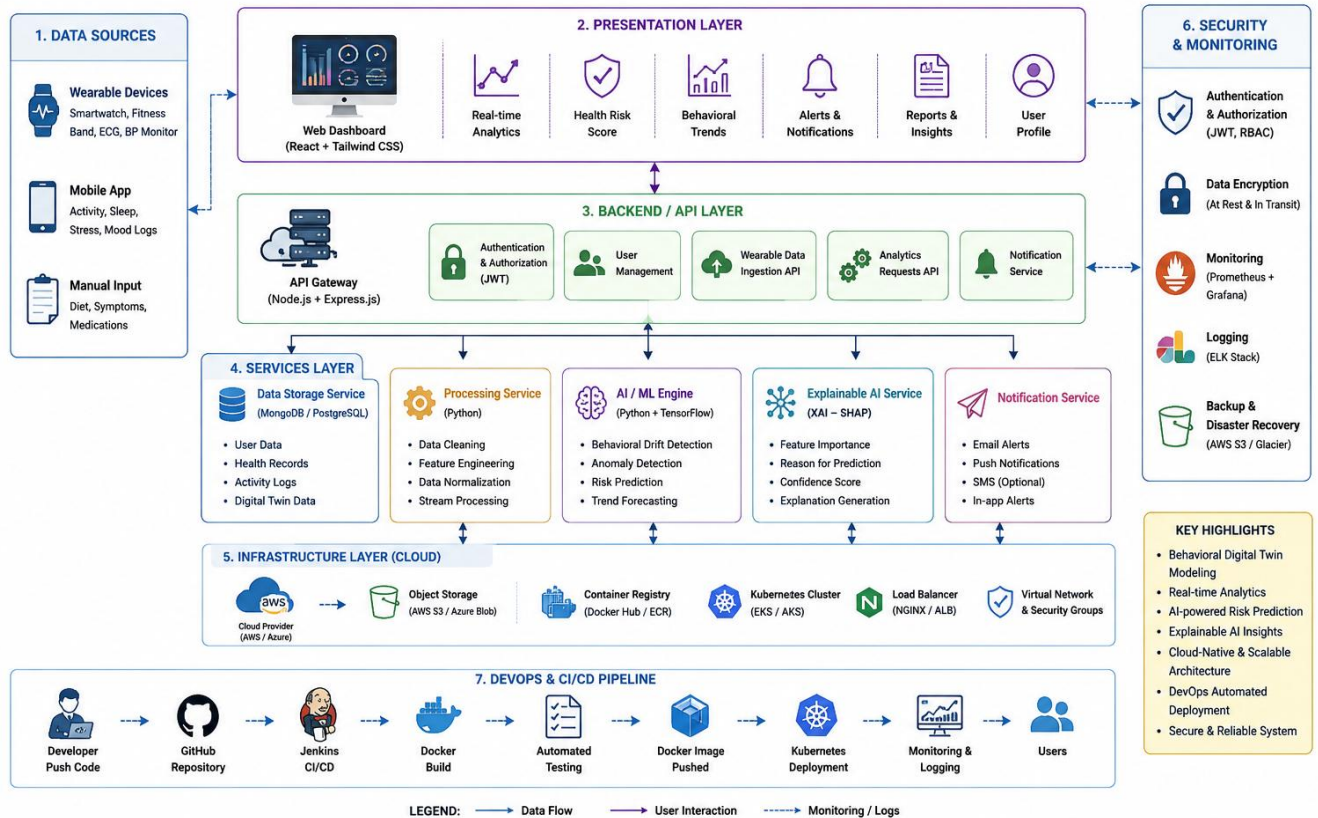
- Kubernetes-based scaling

- Distributed analytics processing
- Cloud-native infrastructure

10. Tech Stack

Layer	Technology	Purpose
Frontend	React + Tailwind CSS	Dashboard UI
Backend	Node.js + Express.js	API Services
AI/ML	Python + TensorFlow + Scikit-learn	Predictive Analytics
Database	MongoDB / PostgreSQL	Health Data Storage
Cloud	AWS / Azure	Cloud Infrastructure
DevOps	Docker + Jenkins + Kubernetes	Deployment Pipeline
Infrastructure	Terraform	Cloud Provisioning
Authentication	JWT	Secure Sessions
Version Control	GitHub	Source Control

Architecture Diagram (Logical)



11. Risk Assessment

Risk	Likelihood	Impact	Mitigation
Inaccurate wearable data	Medium	High	Data preprocessing and filtering
Cloud security threats	Medium	High	Encryption and JWT authentication
AI prediction bias	Medium	High	Model validation and testing
Kubernetes scaling failures	Low	Medium	Auto-scaling configuration
CI/CD deployment errors	Medium	Medium	Automated rollback workflows
Infrastructure misconfiguration	Medium	Medium	Terraform state validation

Model interpretability issues	Medium	High	Explainable AI integration
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